

AI INVESTMENT SIMULATOR

PURPLE CITY

Foundation Study Notes - Portals 1-9

Learn how markets and investing work

Starting capital: \$14,800

Purple City turns the syllabus foundation into an applied learning sequence. The same virtual portfolio continues through every portal: learn, research, decide, allocate, experience, measure, reflect and adapt.

Syllabus-aligned learning guide

The explanations, examples and exercises below expand the Purple City topics into teachable notes while preserving the syllabus progression.

How to Use Purple City

Purple City is not designed as nine isolated lessons. Each portal changes what the learner is able to do with the same \$14,800 portfolio. By the end, the learner should be able to explain an investment objective, understand major asset classes, place simulated orders, interpret basic market information, research evidence, construct a diversified starting portfolio and defend active buy/add/hold/trim/exit decisions.

The Purple City learning loop

- LEARN - understand the concept before risking capital.
- RESEARCH - gather evidence relevant to the decision.
- DECIDE - choose an action and state why.
- ALLOCATE - determine how much capital the idea deserves.
- EXPERIENCE - observe what happens after the decision.
- MEASURE - compare outcome, risk and expectations.
- REFLECT - record what was right, wrong or uncertain.
- ADAPT - revise the portfolio when evidence changes.

Rules for the Portfolio Journal

- Record every transaction and the reason before execution.
- Separate facts from assumptions.
- State what evidence would cause you to change your mind.
- Track cash as an intentional portfolio position, not as “unused money.”
- Judge decisions by process and risk as well as profit or loss.

Portal 1 - Enter the Market

Learning objective

Distinguish investing, saving and trading; explain risk and return, compounding, inflation and real returns.

Portfolio Lab mission

Receive \$14,800 in virtual cash and establish the first investment objective.

1.1 Saving, investing and trading

Saving is primarily about preserving money for near-term needs and emergencies. Investing accepts uncertainty in exchange for the possibility of growing purchasing power over time. Trading focuses more heavily on shorter-term price movements. None is automatically “better”; the correct choice depends on the purpose and time horizon of the money.

Approach	Primary purpose	Typical horizon	Main concern
Saving	Capital preservation / liquidity	Short	Inflation and low return
Investing	Long-term wealth creation	Medium to long	Market and business risk
Trading	Profit from shorter-term moves	Short	Timing, volatility, costs

1.2 Risk and return

Return is the gain or loss generated by an asset. Risk is the possibility that the actual outcome differs from what was expected, including permanent loss of capital. Higher expected returns generally require accepting some form of additional risk, but taking more risk does not guarantee a higher realized return.

1.3 Compound growth

Example

\$14,800 growing at 7% annually would become about \$20,760 after five years if gains were reinvested and there were no taxes or fees. This is an illustration, not a forecast.

Present value (PV) and future value (FV)

Present value is the value today of money that will be received or paid in the future. Future value is what money held today will grow to after earning a return for one or more periods. They are linked by the time value of money: money available today can be invested and earn a return.

$$FV = PV(1 + r)^n$$

PV = present value; FV = future value; r = return or interest rate per period; n = number of compounding periods.

$$PV = FV / (1 + r)^n$$

Present value discounts a future amount back into today's dollars.

Example 1 — Future value for one year

Invest \$1,000 today at 5% for one year.

$$FV = \$1,000(1.05)^1 = \$1,050$$

Example 2 — Compounding for three years

Invest \$2,000 at 8% per year and reinvest each year's gain.

$$FV = \$2,000(1.08)^3 = \$2,519.42$$

Year 1 = \$2,160.00; Year 2 = \$2,332.80; Year 3 = \$2,519.42. Returns are now being earned on earlier returns.

Example 3 — Finding present value

You want \$5,000 in four years. If the investment earns 6% annually, how much is required today?

$$PV = \$5,000 / (1.06)^4 = \$3,960.47$$

Example 4 — Compounding more than once per year

If interest compounds m times per year, the periodic rate is r/m and there are mn periods.

$$FV = PV(1 + r/m)^{mn}$$

For \$3,000 at 6% compounded monthly for two years:

$$FV = \$3,000(1 + 0.06/12)^{24} = \$3,381.48$$

Example 5 — The simulator's \$14,800

If \$14,800 hypothetically earns 7% annually for five years with all gains reinvested:

$$FV = \$14,800(1.07)^5 = \$20,757.57$$

This is a mathematical illustration, not a guaranteed investment outcome.

1.4 Inflation and real return

Nominal return

Nominal return is the percentage gain or loss on an investment measured in current money, before adjusting for inflation. It tells the investor how the dollar value changed, but not whether purchasing power improved.

$$\text{Nominal return} = (\text{Ending value} - \text{Beginning value} + \text{Income}) / \text{Beginning value}$$

Inflation

Inflation is a sustained rise in the general price level of goods and services. As prices rise, each unit of money buys less.

$$\text{Inflation rate} = (\text{Price index}_1 - \text{Price index}_0) / \text{Price index}_0$$

Real return

Real return adjusts nominal investment performance for inflation and therefore measures the change in purchasing power.

$$\text{Real return} = (1 + \text{nominal return}) / (1 + \text{inflation rate}) - 1$$

Why nominal and real returns are both important

Nominal return answers, “How much did my money increase?” Real return answers, “How much did my purchasing power increase?” A positive nominal return can still produce a negative real return when inflation is higher than the investment return. Nominal results matter for account values and cash amounts; real results matter for whether wealth can buy more goods and services.

Example 1 — Positive nominal and real return

Beginning value = \$100.00; ending value = \$108.00; inflation = 3.0%.

$$\text{Nominal return} = (\$108.00 - \$100.00) / \$100.00 = 8.00\%$$

$$\text{Inflation} = (103.0 - 100.0) / 100.0 = 3.00\%$$

$$\text{Real return} = (1 + 0.0800) / (1 + 0.0300) - 1 = 4.85\%$$

Example 2 — Positive nominal but negative real return

Beginning value = \$100.00; ending value = \$104.00; inflation = 6.0%.

$$\text{Nominal return} = (\$104.00 - \$100.00) / \$100.00 = 4.00\%$$

$$\text{Inflation} = (106.0 - 100.0) / 100.0 = 6.00\%$$

$$\text{Real return} = (1 + 0.0400) / (1 + 0.0600) - 1 = -1.89\%$$

Example 3 — Higher return with high inflation

Beginning value = \$5,000.00; ending value = \$5,500.00; inflation = 8.0%.

$$\text{Nominal return} = (\$5,500.00 - \$5,000.00) / \$5,000.00 = 10.00\%$$

$$\text{Inflation} = (108.0 - 100.0) / 100.0 = 8.00\%$$

$$\text{Real return} = (1 + 0.1000) / (1 + 0.0800) - 1 = 1.85\%$$

Example 4 — Low inflation environment

Beginning value = \$2,000.00; ending value = \$2,100.00; inflation = 2.0%.

$$\text{Nominal return} = (\$2,100.00 - \$2,000.00) / \$2,000.00 = 5.00\%$$

$$\text{Inflation} = (102.0 - 100.0) / 100.0 = 2.00\%$$

$$\text{Real return} = (1 + 0.0500) / (1 + 0.0200) - 1 = 2.94\%$$

Example 5 — \$14,800 portfolio

Beginning value = \$14,800.00; ending value = \$15,984.00; inflation = 4.0%.

$$\text{Nominal return} = (\$15,984.00 - \$14,800.00) / \$14,800.00 = 8.00\%$$

$$\text{Inflation} = (104.0 - 100.0) / 100.0 = 4.00\%$$

$$\text{Real return} = (1 + 0.0800) / (1 + 0.0400) - 1 = 3.85\%$$

Portal mission

1. Write one investment objective for the \$14,800.
2. State the intended horizon.
3. State the minimum cash that should remain liquid.
4. Choose a target such as capital growth, income, preservation or a combination.
5. Write one sentence describing what “success” means beyond simply making money.

Portal 2 - Know Your Investor

Learning objective

Understand risk tolerance, risk capacity, investment horizon, liquidity needs and return objectives.

Portfolio Lab mission

Create a personal investment mandate.

2.1 Risk tolerance vs risk capacity

Risk tolerance is psychological: how much uncertainty and loss the investor can emotionally accept. Risk capacity is financial: how much loss the investor can actually afford without jeopardizing essential goals. An investor can have high tolerance but low capacity, or the reverse.

2.2 Time horizon and liquidity

A longer horizon can provide more time to recover from temporary market declines, while a short horizon usually increases the importance of stability and liquidity. Liquidity is the ability to convert an asset to usable cash quickly without accepting a large discount.

2.3 Return objectives

Return objectives should be connected to a purpose and constraint. “I want 20%” is not a complete objective. A better mandate states the goal, time horizon, acceptable risk, liquidity requirement and relevant restrictions.

Investment mandate template

Objective: _____. Horizon: _____. Risk tolerance: low / moderate / high. Risk capacity: low / moderate / high.
Minimum cash reserve: \$ _____. Permitted asset classes: _____. Maximum single-position weight: _____. Review frequency: _____.

Checkpoint questions

- Would a 10% portfolio decline cause you to sell immediately? Here, “sell” means disposing of an investment you own in exchange for cash—for example, selling shares that you previously bought. Why might one investor sell after a decline while another continues holding?
- What portion of the \$14,800 may be needed soon?
- Which matters more for this mandate: avoiding large losses or maximizing growth?
- What is the maximum amount you would allow in one investment?

Portal 3 - Understanding Asset Classes

Learning objective

Recognize stocks, bonds, ETFs, real estate, businesses, forex, commodities and cash.

Portfolio Lab mission

Build an initial asset-class watchlist.

An asset class groups investments that share important economic characteristics. Different assets respond differently to growth, inflation, interest rates, business conditions and investor sentiment. Understanding these differences is the foundation of diversification.

Asset class	What you own / gain exposure to	Potential return source	Key risks
Stocks	Equity in public companies	Earnings growth, dividends, valuation	Business, market, valuation
Bonds	Debt claim	Interest and principal repayment	Rates, inflation, credit
ETFs	Basket of assets	Depends on holdings	Market, tracking, concentration
Real estate	Property / REIT exposure	Rent, appreciation	Vacancy, rates, leverage, illiquidity
Businesses	Whole operating company	Cash flow and enterprise growth	Execution, financing, concentration
Forex	Relative value of currencies	Exchange-rate movement, carry	Macro, leverage, policy
Commodities	Physical-resource exposure	Price movement	Cycles, volatility, storage/roll effects
Cash	Liquid currency / equivalents	Interest	Inflation, opportunity cost

How major asset classes tend to behave

Stocks

Ownership in companies. Prices respond to expected profits, economic growth, interest rates, company news and investor expectations.

Bonds

Loans to governments or companies. Bond prices are affected by interest rates, inflation expectations, credit quality and maturity. When market interest rates rise, existing fixed-rate bond prices generally fall; when rates fall, they generally rise. Deteriorating credit quality can also reduce bond prices.

ETFs

Exchange-traded funds are pooled investment vehicles whose shares trade on an exchange. An ETF may hold many stocks, bonds, commodities or other assets. Many track an index. Examine its holdings, objective, fees/expense ratio, liquidity, tracking difference, concentration and currency exposure.

Real estate

Can generate rental income and appreciation. Examine location, rents, occupancy/vacancy, operating expenses, financing and mortgage rates, local supply, employment/population trends, taxes, maintenance, property condition and comparable sales. Direct property is relatively illiquid; REITs provide exchange-traded property exposure.

Cash

Provides liquidity and stability but usually has lower expected long-term returns; inflation can erode purchasing power.

Commodities

Oil, gas, gold, agricultural products and others. Prices are influenced by global supply/demand, inventories, weather, geopolitics, production disruptions, inflation expectations and currencies.

Forex

Currencies trade in pairs. Exchange rates respond to interest-rate differentials, inflation, economic growth, central-bank policy, trade flows, political risk and expectations.

Private businesses

Ownership in operating companies that may be hard to value or sell. Consider revenue, cash flow, competitive advantage, management, customers, debt and exit possibilities.

Watchlist exercise

Choose at least one example from six asset classes. For each, record: what it is, why it might belong in a portfolio, its main risk, what information would move its value, and whether it is currently a candidate for investment or only observation.

Portal 4 - How Markets Work

Learning objective

Understand exchanges, brokers, bid/ask prices, liquidity, order types and transaction costs.

Portfolio Lab mission

Execute the first simulated transaction.

4.1 Market infrastructure

An exchange is a venue where eligible securities are bought and sold. A broker provides access to markets and handles orders. Buyers submit bids; sellers submit asks. The bid-ask spread is the difference between the best available buying and selling prices and is one form of trading friction.

4.2 Liquidity

A liquid asset can generally be traded quickly with limited impact on price. Thinly traded assets can have wider spreads and greater slippage, meaning the actual execution price differs from the price the investor expected.

4.3 Market vs limit orders

- Market order: prioritizes execution, but not a specific price.
- Limit order: specifies the worst acceptable price, but execution is not guaranteed.

4.4 Transaction costs

Costs include explicit commissions or fees and implicit costs such as spreads, slippage and market impact. Frequent activity can erode returns even when individual decisions appear small.

First trade checklist

What am I buying? Why now? How much? What order type is appropriate? What is the estimated cost? What would make me exit? What percentage of the total \$14,800 will this position represent?

4.1 Market infrastructure — in detail

A financial market connects buyers and sellers. For exchange-traded securities, an investor normally sends an order through a broker. The broker routes the order to an exchange, market maker or another venue. The market matches buying interest with selling interest.

The bid is the highest current price a buyer is willing to pay. The ask is the lowest current price a seller will accept. If the best bid is \$49.95 and the best ask is \$50.05, the spread is \$0.10.

4.2 What creates liquidity?

Liquidity is supported by many willing buyers and sellers, frequent trading, substantial quantities available near the current price, accessible information and market-making activity. It can deteriorate when uncertainty rises, participants withdraw, an asset is obscure, or a shock creates one-sided trading.

Practical signs of liquidity include narrow bid-ask spreads, meaningful volume, many orders close to the current price, and the ability to trade without moving the price much.

4.3 Limit orders and interaction with the market

A market order prioritizes execution at the best prices currently available. A limit order prioritizes price: a buy limit is the maximum price the investor will pay, while a sell limit is the minimum price the investor will accept.

Example: ABC has a bid of \$49.90 and ask of \$50.10. A market buy normally begins executing against sellers at or near \$50.10. A buy limit at \$49.50 waits because current sellers want more; it can execute only if a seller becomes willing to transact at \$49.50 or less. A buy limit at \$50.10 may execute immediately if sufficient shares are available.

Limit orders give price control but may never execute. Market orders increase the chance of execution but can produce an unexpectedly poor price in a fast or illiquid market.

4.4 Transaction-cost examples

Example 1 — Spread: Buy 100 shares at the \$20.10 ask and immediately sell at the \$20.00 bid. Spread cost = $100 \times \$0.10 = \10 .

Example 2 — Commission: \$5 to buy plus \$5 to sell = \$10 total explicit commission.

Example 3 — Slippage: You expect 200 shares at \$30.00 but receive an average execution price of \$30.08. Slippage = $200 \times \$0.08 = \16 .

Example 4 — Market impact: A large buy order may consume the cheapest available sell orders and then execute against more expensive offers, raising the average purchase price.

Portal 5 - Reading the Market

Learning objective

Interpret price charts, market capitalization, indices, trading volume, volatility and sentiment.

Portfolio Lab mission

Interpret basic market information before making a decision.

5.1 Price charts

Charts summarize historical price behavior. They can reveal direction, ranges, gaps and periods of instability, but a chart does not explain why an asset is worth its price. Historical patterns are evidence about behavior, not guarantees about the future.

5.2 Market capitalization

For a public company, market capitalization is approximately share price multiplied by shares outstanding. It represents the market value of the company's equity, not necessarily the value of the entire business including debt.

5.3 Indices

An index tracks a defined group of securities and can serve as a market barometer or benchmark. A portfolio should be compared with a benchmark that reflects the type of risk the learner is actually taking.

5.4 Volume, volatility and sentiment

- Volume measures how much of an asset changes hands over a period.
- Volatility describes the magnitude of price fluctuations; it is a risk indicator, not a complete definition of risk.
- Sentiment reflects how optimistic or pessimistic market participants appear to be. Sentiment can influence price without changing fundamental value.

Market-reading worksheet

- Current price and recent range
- Market capitalization or asset size where relevant
- Relevant benchmark/index
- Recent volume compared with normal activity
- Observed volatility
- One possible fundamental explanation
- One possible sentiment-driven explanation

5.1 Worked example: reading a price chart

This fictional share rises from \$100 in January to \$127 in December, with temporary declines. A beginner should first describe direction, highs/lows and variability before attempting to explain why the price changed.

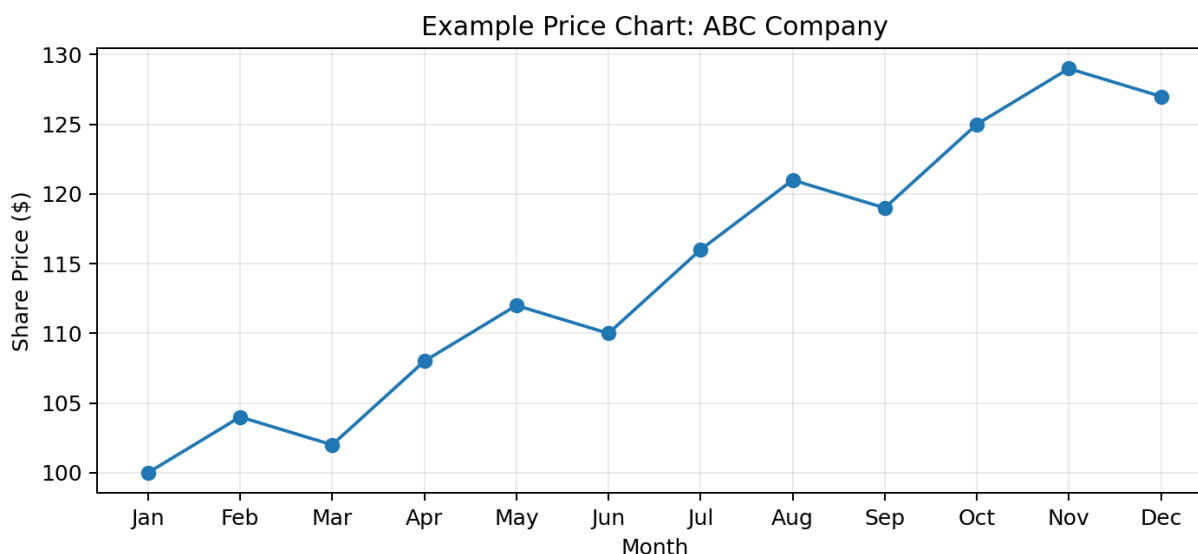


Figure 1. Fictional ABC Company monthly price series for teaching purposes.

Line charts commonly connect closing prices. Bar/OHLC and candlestick charts can show open, high, low and close. Charts summarize historical behavior; they do not establish fundamental value.

5.2 Market capitalization — calculation and interpretation

Market capitalization (market cap) is the market value of a public company's outstanding equity shares.

$$\text{Market capitalization} = \text{Current share price} \times \text{Shares outstanding}$$

Example 1: 10 million shares $\times$ \$25 = \$250 million market cap.

Example 2: 500 million shares $\times$ \$80 = \$40 billion market cap.

Example 3: Company A trades at \$200 with 1 million shares = \$200 million market cap. Company B trades at \$20 with 50 million shares = \$1 billion market cap. A higher share price does not necessarily mean a larger company.

Market cap changes with share price and with shares outstanding. It measures equity value, not the total value of the entire enterprise including debt and cash.

5.3 Market indices — examples and uses

An index is a rules-based measure representing a defined group of securities. Investors use indices as market barometers and benchmarks for judging portfolio performance.

S&P 500 — 500 leading U.S. companies; a common large-cap U.S. equity benchmark.

Dow Jones Industrial Average — 30 prominent U.S. companies; price-weighted.

Nasdaq Composite — thousands of Nasdaq-listed securities, with substantial technology exposure.

FTSE 100 — 100 large companies listed on the London Stock Exchange.

MSCI World — large- and mid-cap equities across developed markets; useful as a developed-global-equity benchmark.

Indices differ in membership and weighting. A market-cap-weighted index gives larger companies more influence; a price-weighted index gives higher-priced shares more influence. Choose a benchmark that resembles the portfolio's investment universe and risk.

5.4 Volume, volatility and sentiment — developed

Volume measures how many shares, contracts or units trade during a period. Unusually high volume can indicate unusually strong participation in a price move, but volume does not tell you whether an asset is fundamentally good value.

Volatility describes the magnitude and variability of price or return movements. High volatility means prices have historically moved more dramatically. It is a useful risk indicator but is not identical to the risk of permanent capital loss.

Sentiment is the prevailing optimism, pessimism or risk appetite of market participants. It can be inferred imperfectly from surveys, fund flows, options activity, news tone, positioning and price behavior.

The key distinction is fundamental information versus market reaction. A strong earnings report is fundamental evidence. The price reaction depends partly on what investors expected beforehand. A company can report higher profits and still fall if the market expected even stronger results.

$$\text{Sample volatility of returns: } s = \sqrt{[\Sigma(r_i - \bar{r})^2 / (n - 1)]}$$

Worked interpretation example

ABC rises 8% in one day, volume is three times normal, and the company announces earnings above expectations plus higher future guidance. The price move, volume and sentiment are informative, but the investor should still examine valuation, sustainability of earnings and whether the good news was already reflected in the price.

Portal 6 - Research Fundamentals

Learning objective

Use company reports, economic data, industry research and news; judge source credibility and maintain research logs.

Portfolio Lab mission

Research several opportunities and rank the quality of the evidence.

6.1 Evidence hierarchy

Good investment research distinguishes primary evidence from commentary. Company filings, audited reports, official economic releases and regulatory disclosures are often closer to the underlying facts. Analyst reports, journalism and expert commentary can add interpretation. Social media can reveal sentiment or leads, but claims should be verified before they influence capital allocation.

6.2 Source credibility test

- Authority - Who produced the information and what expertise do they have?
- Evidence - Are claims supported by data or primary documents?
- Timeliness - Is the information current enough for the decision?
- Incentives - Does the source benefit if you believe the claim?

- Consistency - Does it agree with independent credible evidence?
- Specificity - Can the claim be tested or is it vague?

6.3 Research log

Date	Asset	Question	Source	Finding	Decision impact

Research discipline

Do not begin with “Why should I buy this?” Begin with “What would I need to know to decide whether this deserves capital?” This reduces confirmation bias and forces the learner to search for disconfirming evidence.

Portal 7 - Build Your First Portfolio

Learning objective

Understand position sizing, portfolio weights, diversification and cash allocation.

Portfolio Lab mission

Deploy part or all of the \$14,800 into a defensible starting portfolio.

7.1 Position size and portfolio weight

Position size is the dollar amount committed to an investment. Portfolio weight is that position as a percentage of the entire portfolio. Position sizing is a risk decision: even an excellent idea can damage a portfolio if the exposure is too large.

Portfolio weight = position value / total portfolio value x 100%.

Example allocation

Illustrative only: \$5,180 stocks (35%), \$2,960 bonds (20%), \$2,220 ETF (15%), \$1,480 real-estate exposure (10%), \$740 commodity exposure (5%) and \$2,220 cash (15%) = \$14,800. The purpose is to practice allocation, not prescribe a “correct” portfolio.

7.2 Diversification

Diversification spreads exposure across investments whose risks are not identical. Owning many securities is not automatically diversified: ten companies from the same industry can still represent one concentrated economic bet.

7.3 Cash allocation

Cash provides liquidity, reduces short-term volatility and preserves optionality for future opportunities. Its cost is that long periods in cash can lose purchasing power to inflation and may underperform productive assets.

Portfolio construction steps

6. Re-read the mandate from Portal 2.
7. Choose target asset-class weights.
8. Select candidate investments supported by Portal 6 research.

9. Set maximum single-position size.
10. Decide how much cash to retain.
11. Check concentration by asset class, sector, geography and currency.
12. Execute simulated orders and record the thesis for each.

Portal 8 - Investment Decisions

Learning objective

Practice buy, add, hold, trim and exit decisions; understand opportunity cost and decision rationale.

Portfolio Lab mission

Make and document active portfolio decisions.

8.1 The five active decisions

Decision	Meaning	Typical justification
BUY	Open a new position	Expected reward justifies risk
ADD	Increase an existing position	Evidence strengthens or valuation improves
HOLD	Keep exposure unchanged	Thesis remains intact and size is appropriate
TRIM	Reduce exposure	Valuation, concentration or risk has risen
EXIT	Close the position	Thesis breaks or capital has a superior use

8.2 Opportunity cost

Capital committed to one investment cannot simultaneously be used elsewhere. A holding should therefore compete not only against cash, but against the best alternative opportunity available at similar risk.

8.3 Decision rationale

A strong rationale states the action, evidence, expected outcome, principal risks, position size and conditions for reassessment. “The price is going up” is an observation, not a complete investment thesis.

Decision journal prompt

Action: BUY / ADD / HOLD / TRIM / EXIT. Evidence: _____. What changed: _____. Expected reward: _____. Main risk: _____. New portfolio weight: _____. What would prove this decision wrong: _____. Review date/event: _____.

Portal 9 - Purple City Challenge

Learning objective

Integrate the foundation: holdings, allocation, evidence, risk and portfolio weaknesses.

Portfolio Lab mission

Defend the initial portfolio and rebalance before entering Blue City.

9.1 The challenge

The Purple City Challenge is a defense of process, not a contest to see who earned the highest short-term return. The learner should be able to explain why each holding exists, why its size is appropriate, what evidence supports it, how it fits the mandate and what could cause the position to change.

9.2 Portfolio defense structure

13. Mandate - objective, horizon, risk profile and liquidity needs.
14. Allocation - asset-class weights and cash reserve.
15. Holdings - purpose and thesis for every position.
16. Evidence - strongest sources used and important uncertainties.
17. Risk - largest concentrations and plausible loss scenarios.
18. Decision quality - explain at least one buy/add/hold/trim/exit decision.
19. Weaknesses - identify what is missing or poorly supported.
20. Rebalance - make changes required before Blue City.

9.3 Purple City scoring rubric

Area	What strong work demonstrates	Weight
Investment mandate	Clear objective and constraints	15%
Research quality	Credible evidence and research log	20%
Portfolio construction	Defensible sizing, cash and diversification	20%
Decision rationale	Actions linked to evidence and opportunity cost	20%
Risk awareness	Concentrations and downside recognized	15%
Reflection	Mistakes/uncertainty identified and portfolio adapted	10%

Purple City Quick Reference

Core vocabulary

Term	Meaning
Nominal return	Return measured in money terms before adjusting for inflation.
Real return	Return after accounting for inflation.
Liquidity	Ability to convert an asset into cash quickly with limited price concession.
Bid	Highest current price a buyer is offering.
Ask	Lowest current price a seller is offering.
Spread	Difference between bid and ask.
Volatility	Magnitude of price fluctuations.
Market capitalization	Market value of a public company's equity.
Diversification	Spreading exposure across non-identical sources of risk.
Position size	Dollar value committed to one investment.
Portfolio weight	Position value as a percentage of total portfolio value.
Opportunity cost	Value of the best alternative use of capital.

Before unlocking Blue City

- I can explain the difference between saving, investing and trading.
- I can state my investment mandate and constraints.
- I can describe the main asset classes and their risks.
- I understand basic market mechanics and order types.
- I can interpret basic market data without treating charts as certainty.
- I can rank sources by credibility and maintain a research log.
- I can calculate and defend position weights.
- I can distinguish BUY, ADD, HOLD, TRIM and EXIT.
- I can defend and rebalance my \$14,800 portfolio based on evidence.

Purple City completion standard

The learner is ready for Blue City when the portfolio is defensible even if its short-term performance is disappointing. The next stage shifts from “How does investing work?” to “What should I invest in, and what is it worth?”